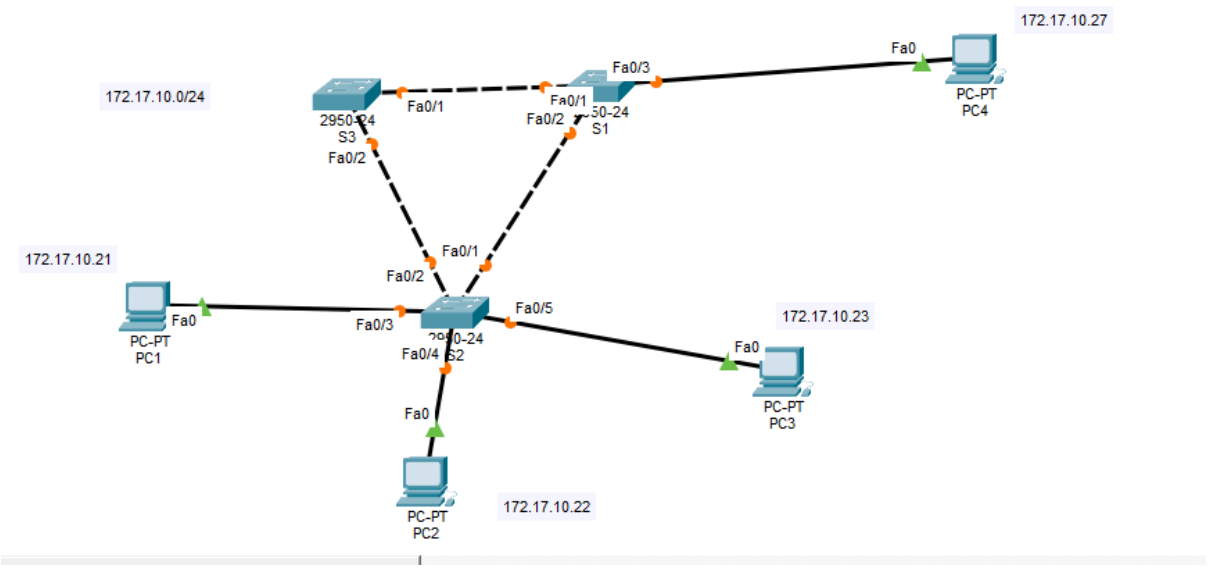
LAB2 - Vlan -STP

VLAN and Spanning Tree .....
1. Create VLAN
2. VTP modes
3. Configuration of the Spanning Tree

Device	Interface	IP address	Subnet mask	Default gateway
S1	VLAN1	172.17.10.1	255.255.255.0	N/A
S2	VLAN1	172.17.10.2	255.255.255.0	N/A
S3	VLAN1	172.17.10.3	255.255.255.0	N/A
PC1	NIC	172.17.10.21	255.255.255.0	172.17.10.1
PC2	NIC	172.17.10.22	255.255.255.0	172.17.10.1
PC3	NIC	172.17.10.23	255.255.255.0	172.17.10.1
PC4	NIC	172.17.10.27	255.255.255.0	172.17.10.1

Step 1: Topologie



Step 2: Configure Host PCs and verify the switch configuration
Les postes PC1, PC2, PC3 et PC4 sont configurés avec l'adresse IP, le masque de sous-réseau et la passerelle indiqués dans le tableau d'adressage fourni au début du laboratoire.

IP Configuration

Interface: FastEthernet0

IP Configuration

DHCP | Static

IPv4 Address: 172.17.10.27

Subnet Mask: 255.255.255.0

Default Gateway: 172.17.10.1

DNS Server: 0.0.0.0

IP Configuration

Interface: FastEthernet0

IP Configuration

DHCP | Static

IPv4 Address: 172.17.10.23

Subnet Mask: 255.255.255.0

Default Gateway: 172.17.10.1

DNS Server: 0.0.0.0

IP Configuration

Interface: FastEthernet0

IP Configuration

DHCP | Static

IPv4 Address: 172.17.10.22

Subnet Mask: 255.255.255.0

Default Gateway: 172.17.10.1

DNS Server: 0.0.0.0

IP Configuration

Interface: FastEthernet0

IP Configuration

DHCP | Static

IPv4 Address: 172.17.10.21

Subnet Mask: 255.255.255.0

Default Gateway: 172.17.10.1

DNS Server: 0.0.0.0

IPv6 Configuration

Step 3: Création du vlan 1 sur les 3 switches

Nous allons attribuer une adresse IP de management aux trois switches S1, S2 et S3. Cette adresse, configurée sur l'interface virtuelle VLAN 1, permet d'administrer les équipements à distance et de vérifier leur accessibilité au sein du réseau.

```
YG-S1#conf
YG-S1#configure t
YG-S1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
YG-S1(config)#int
YG-S1(config)#interface vla
YG-S1(config)#interface vlan 1
YG-S1(config-if)#?
Interface configuration commands:
  arp           Set arp type (arpa, probe, snap) or timeout
  description   Interface specific description
  exit          Exit from interface configuration mode
  ip            Interface Internet Protocol config commands
  no            Negate a command or set its defaults
  shutdown      Shutdown the selected interface
  standby       HSRP interface configuration commands
YG-S1(config-if)#ip ?
  address         Set the IP address of an interface
  helper-address  Specify a destination address for UDP broadcasts
YG-S1(config-if)#ip ad
YG-S1(config-if)#ip address ?
  A.B.C.D  IP address
  dhcp     IP Address negotiated via DHCP
YG-S1(config-if)#ip address 172.17.10.1 ?
  A.B.C.D  IP subnet mask
YG-S1(config-if)#ip address 172.17.10.1 255.255.255.0?
A.B.C.D
YG-S1(config-if)#ip address 172.17.10.1 255.255.255.0 ?
  <cr>
YG-S1(config-if)#ip address 172.17.10.1 255.255.255.0
YG-S1(config-if)#ex
YG-S1(config-if)#exit
YG-S1(config)#
```

```
Switch>en
Switch>enable
Switch#conf
Switch#configure t
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#ho
Switch(config)#hostname YG-S2
YG-S2(config)#int
YG-S2(config)#interface vl
YG-S2(config)#interface vlan 1 ?
  <cr>
YG-S2(config)#interface vlan 1
YG-S2(config-if)#ip
YG-S2(config-if)#ip ad
YG-S2(config-if)#ip address 172.17.10.2 255.255.255.0
YG-S2(config-if)#ex
YG-S2(config)#
```

```
Switch>en
Switch#conf
Switch#configure t
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#h
Switch(config)#hostname YG-S3
YG-S3(config)#int
YG-S3(config)#interface v
YG-S3(config)#interface vlan 1
YG-S3(config-if)#ip
YG-S3(config-if)#ip ad
YG-S3(config-if)#ip address 172.17.10.3 255.255.255.0
YG-S3(config-if)#ex
YG-S3(config-if)#exit
YG-S3(config)#
```

Step 4: Enable the user ports on S1 and S2

En se référant au schéma de topologie fourni dans le laboratoire, identifions les ports des switches S1 et S2 qui sont connectés aux équipements utilisateurs. Sur S2, trois ports sont dédiés à la connexion des postes finaux et sur S1 un seul port dédié aux équipements finaux. Ces ports doivent être configurés pour accepter uniquement du trafic provenant d'un seul VLAN, ce qui correspond au mode d'accès.

```
Switch(config-if-range)#ex
Switch(config)#h
Switch(config)#hostname YG-S2
YG-S2(config)#interface range fa0/3, fa0/4, fa0/5
YG-S2(config-if-range)#switchport mode access
YG-S2(config-if-range)#no shutdown
YG-S2(config-if-range)#ex
YG-S2(config)#
```

```
Enter configuration commands, one per line.  End with CNTL/Z.
YG-S1(config)#int
YG-S1(config)#interface vla
YG-S1(config)#interface vlan 1
YG-S1(config-if)#?
Interface configuration commands:
  arp           Set arp type (arpa, probe, snap) or timeout
  description   Interface specific description
  exit          Exit from interface configuration mode
  ip            Interface Internet Protocol config commands
  no            Negate a command or set its defaults
  shutdown      Shutdown the selected interface
  standby       HSRP interface configuration commands
YG-S1(config-if)#ip ?
  address          Set the IP address of an interface
  helper-address   Specify a destination address for UDP broadcasts
YG-S1(config-if)#ip ad
YG-S1(config-if)#ip address ?
  A.B.C.D  IP address
  dhcp     IP Address negotiated via DHCP
YG-S1(config-if)#ip address 172.17.10.1 ?
  A.B.C.D  IP subnet mask
YG-S1(config-if)#ip address 172.17.10.1 255.255.255.0?
A.B.C.D
YG-S1(config-if)#ip address 172.17.10.1 255.255.255.0 ?
  <cr>
YG-S1(config-if)#ip address 172.17.10.1 255.255.255.0
YG-S1(config-if)#ex
YG-S1(config-if)#exit
YG-S1(config)#int
YG-S1(config)#interface ra
YG-S1(config)#interface range f0/3
YG-S1(config-if-range)#sw
YG-S1(config-if-range)#switchport mo
YG-S1(config-if-range)#switchport mode a
YG-S1(config-if-range)#switchport mode access
YG-S1(config-if-range)#no
YG-S1(config-if-range)#no sh
YG-S1(config-if-range)#no shutdown
YG-S1(config-if-range)#
```

```
YG-S1#show interfaces fastEthernet 0/3 switchport
Name: Fa0/3
Switchport: Enabled
Administrative Mode: static access
Operational Mode: static access
Administrative Trunking Encapsulation: dot1q
Operational Trunking Encapsulation: native
Negotiation of Trunking: Off
Access Mode VLAN: 1 (default)
Trunking Native Mode VLAN: 1 (default)
Voice VLAN: none
Administrative private-vlan host-association: none
Administrative private-vlan mapping: none
Administrative private-vlan trunk native VLAN: none
Administrative private-vlan trunk encapsulation: dot1q
Administrative private-vlan trunk normal VLANs: none
Administrative private-vlan trunk private VLANs: none
Operational private-vlan: none
Trunking VLANs Enabled: All
Pruning VLANs Enabled: 2-1001
Capture Mode Disabled
Capture VLANs Allowed: ALL
Protected: false
Appliance trust: none


YG-S1#
```

Step 5: Enable trunk ports on S1, S2, and S3
Cette étape consiste à activer le mode trunk sur l'ensemble des liens reliant les switches S1, S2 et S3. Ce mode (trunking) permet de transporter simultanément plusieurs VLANs sur une même liaison, ce qui rend la topologie évolutive et compatible avec l'ajout de nouveaux VLANs dans les étapes suivantes du laboratoire.

```
YG-S1(config)#interface range fa0/1, fa0/2
YG-S1(config-if-range)#mo
YG-S1(config-if-range)#sw
YG-S1(config-if-range)#switchport mo
YG-S1(config-if-range)#switchport mode tr
YG-S1(config-if-range)#switchport mode trunk
YG-S1(config-if-range)#no sh
YG-S1(config-if-range)#no shutdown
YG-S1(config-if-range)#ex
YG-S1(config-if-range)#exit
YG-S1(config)#sh
YG-S1(config)#sho
YG-S1(config)#ex
YG-S1#
%SYS-5-CONFIG_I: Configured from console by console

YG-S1#sh
YG-S1#show in
YG-S1#show interfaces tr
YG-S1#show interfaces trunk
Port        Mode         Encapsulation  Status        Native vlan
Fa0/1       on           802.1q         trunking      1
Fa0/2       on           802.1q         trunking      1

Port        Vlans allowed on trunk
Fa0/1       1-1005
Fa0/2       1-1005

Port        Vlans allowed and active in management domain
Fa0/1       1
Fa0/2       1

Port        Vlans in spanning tree forwarding state and not pruned
Fa0/1       1
Fa0/2       1

YG-S1#
```

```
YG-S3>en
YG-S3#sh
YG-S3#show in
YG-S3#show interfaces tr
YG-S3#show interfaces trunk
Port        Mode         Encapsulation  Status        Native vlan
Fa0/1       on           802.1q         trunking      1
Fa0/2       on           802.1q         trunking      1

Port        Vlans allowed on trunk
Fa0/1       1-1005
Fa0/2       1-1005

Port        Vlans allowed and active in management domain
Fa0/1       1,10,20,30
Fa0/2       1,10,20,30

Port        Vlans in spanning tree forwarding state and not pruned
Fa0/1       1,10,30
Fa0/2       1,10,20,30

YG-S3#
```

```
YG-S2>
YG-S2>en
YG-S2#sh
YG-S2#show in
YG-S2#show interfaces tr
YG-S2#show interfaces trunk
Port        Mode         Encapsulation  Status        Native vlan
Fa0/1       on           802.1q         trunking      1
Fa0/2       on           802.1q         trunking      1

Port        Vlans allowed on trunk
Fa0/1       1-99
Fa0/2       1-99

Port        Vlans allowed and active in management domain
Fa0/1       1,10,20,30
Fa0/2       1,10,20,30

Port        Vlans in spanning tree forwarding state and not pruned
Fa0/1       1,10,20,30
Fa0/2       20

YG-S2#
```

Exemple du ping

```
Sending 5, 100-byte ICMP Echos to 172.17.10.3, timeout is 2 seconds:
..!!!
Success rate is 60 percent (3/5), round-trip min/avg/max = 0/0/0 ms

YG-S1#ping 172.17.10.3

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 172.17.10.3, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/0 ms

YG-S1#
```

Step 6: Create a new VLANS on S2 with the id 10, 20 and 30 with names finance, staff and guest

Nous allons créer trois nouveaux VLANs sur le switch S2 et leur attribuer des noms correspondant aux différents services du réseau. Une fois créés, ces VLANs doivent être associés aux ports utilisateurs appropriés afin de segmenter logiquement le réseau selon les besoins fonctionnels.

```
YG-S2(config-if-range)#switchport acc
YG-S2(config-if-range)#switchport access vl
YG-S2(config-if-range)#switchport access vlan 20
YG-S2(config-if-range)#ex
YG-S2(config)#vl
YG-S2(config)#vlan 30
YG-S2(config-vlan)#na
YG-S2(config-vlan)#name guest
YG-S2(config-vlan)#ex
YG-S2(config)#int
YG-S2(config)#interface ran
YG-S2(config)#interface range f0/4-15
YG-S2(config-if-range)#sw
YG-S2(config-if-range)#switchport a
YG-S2(config-if-range)#switchport access vl
YG-S2(config-if-range)#switchport access vlan 30
YG-S2(config-if-range)#ex
YG-S2(config)#ex
YG-S2#
%SYS-5-CONFIG_I: Configured from console by console

YG-S2#
YG-S2#sh
YG-S2#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/21, Fa0/22, Fa0/23, Fa0/24
10   finance                          active    Fa0/16, Fa0/17, Fa0/18, Fa0/19
                                                Fa0/20
20   staff                            active    Fa0/3
30   guest                            active    Fa0/4, Fa0/5, Fa0/6, Fa0/7
                                                Fa0/8, Fa0/9, Fa0/10, Fa0/11
                                                Fa0/12, Fa0/13, Fa0/14, Fa0/15
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
YG-S2#
```

Ping from PC3 the PC1, PC2 and PC4. Comment the obtained result.

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 172.17.10.21

Pinging 172.17.10.21 with 32 bytes of data:

Reply from 172.17.10.21: bytes=32 time=15ms TTL=128
Reply from 172.17.10.21: bytes=32 time<1ms TTL=128
Reply from 172.17.10.21: bytes=32 time<1ms TTL=128
Reply from 172.17.10.21: bytes=32 time<1ms TTL=128

Ping statistics for 172.17.10.21:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 15ms, Average = 3ms

C:\>ping 172.17.10.21

Pinging 172.17.10.21 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 172.17.10.21:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```

La capture montre bien deux ping, lors du premier ping il y a succès car il y avait pas d'autres Vlans sur S2 que vlan 1. Mais après avoir créé les vlans 10, 20, 30 sur S2 le ping vers PC1 n'a pas abouti car typiquement ces deux PCs sont pas sur le même réseau les interfaces ont été assignés au différents vlans, elles sont partagés entre les vlans.

Step 7: VTP

Cette partie est intéressante, l'objectif est de mettre en place le **VLAN Trunking Protocol (VTP)** afin de faciliter le partage et la synchronisation de la base de VLANs entre les switches du réseau. VTP permet d'éviter une configuration manuelle répétitive sur chaque switch et garantit une cohérence globale de la topologie. Mais nous allons manipuler quelques commandes pour démontrer comment définir certaines choses si on opte pour un autre chemin.

Utilisons "un show vtp status", alors on voit que sur les switches le mode vtp par défaut est "server"

```
  status    vtp domain status
YG-S2#show vtp st
YG-S2#show vtp status
VTP Version                     : 1
Configuration Revision          : 6
Maximum VLANs supported locally : 255
Number of existing VLANs        : 8
VTP Operating Mode              : Server
VTP Domain Name                 :
VTP Pruning Mode                : Disabled
VTP V2 Mode                     : Disabled
VTP Traps Generation            : Disabled
MD5 digest                      : 0x26 0x5B 0x52 0x09 0xA3 0xF8 0x8A 0xE9
Configuration last modified by 172.17.10.2 at 3-1-93 01:41:20
Local updater ID is 172.17.10.2 on interface Vl1 (lowest numbered VLAN interface found)
YG-S2#
```

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1-Set the name of VTP domain “My office” and password “cisco” of S2

```
YG-S2#conf
YG-S2#configure t
YG-S2#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
YG-S2(config)#vt
YG-S2(config)#vtp do
YG-S2(config)#vtp domain ?
  WORD  The ascii name for the VTP administrative domain.
YG-S2(config)#vtp domain My-office ?
  <cr>
YG-S2(config)#vtp domain My-office
Changing VTP domain name from NULL to My-office
YG-S2(config)#%SW_VLAN-6-VTP_DOMAIN_NAME_CHG: VTP domain name changed to My-office.

YG-S2(config)#vtp pas
YG-S2(config)#vtp password cisco
Setting device VLAN database password to cisco
YG-S2(config)#ex
YG-S2#
%SYS-5-CONFIG_I: Configured from console by console

YG-S2#show vtp status
VTP Version                     : 1
Configuration Revision          : 0
Maximum VLANs supported locally : 255
Number of existing VLANs        : 8
VTP Operating Mode              : Server
VTP Domain Name                 : My-office
VTP Pruning Mode                : Disabled
VTP V2 Mode                     : Disabled
VTP Traps Generation            : Disabled
MD5 digest                      : 0xA5 0x03 0xE1 0x29 0xB6 0x05 0x39 0x2A
Configuration last modified by 172.17.10.2 at 3-1-93 01:41:20
Local updater ID is 172.17.10.2 on interface Vl1 (lowest numbered VLAN interface found)
YG-S2#
```

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2-Change the vtp mode for S1 to client, with the same domain and password of the server, then report the command line.

```
Enter configuration commands, one per line.  End with CNTL/Z.
YG-S1(config)#vt
YG-S1(config)#vtp ?
  domain    Set the name of the VTP administrative domain.
  mode      Configure VTP device mode
  password  Set the password for the VTP administrative domain
  version   Set the adminstrative domain to VTP version
YG-S1(config)#vtp mo
YG-S1(config)#vtp mode ?
  client       Set the device to client mode.
  server       Set the device to server mode.
  transparent  Set the device to transparent mode.
YG-S1(config)#vtp mode client ?
  <cr>
YG-S1(config)#vtp mode client
Setting device to VTP CLIENT mode.
YG-S1(config)#ex
YG-S1#
%SYS-5-CONFIG_I: Configured from console by console

YG-S1#show vtp status
VTP Version                     : 1
Configuration Revision          : 0
Maximum VLANs supported locally : 255
Number of existing VLANs        : 5
VTP Operating Mode              : Client
VTP Domain Name                 : My-office
VTP Pruning Mode                : Disabled
VTP V2 Mode                     : Disabled
VTP Traps Generation            : Disabled
MD5 digest                      : 0xE2 0xD6 0x39 0x2B 0xFC 0x17 0x29 0x9B
Configuration last modified by 0.0.0.0 at 0-0-00 00:00:00
YG-S1#conf
YG-S1#configure t
YG-S1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
YG-S1(config)#vtp pa
YG-S1(config)#vtp password cisco
Setting device VLAN database password to cisco
YG-S1(config)#
```

```
VTP Version                     : 1
Configuration Revision          : 2
Maximum VLANs supported locally : 255
Number of existing VLANs        : 9
VTP Operating Mode              : Client
VTP Domain Name                 : My-office
VTP Pruning Mode                : Disabled
VTP V2 Mode                     : Disabled
VTP Traps Generation            : Disabled
MD5 digest                      : 0x30 0xE1 0x84 0x93 0xD8 0xB2 0x08 0x64
Configuration last modified by 172.17.10.2 at 3-1-93 03:26:36
YG-S1#conf
YG-S1#configure t
YG-S1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
YG-S1(config)#vla
YG-S1(config)#vlan 2
YG-S1(config)#vlan 2
VTP VLAN configuration not allowed when device is in CLIENT mode.
YG-S1(config)#
```

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3- Try to create a new Vlan on the S2. Is it possible? explique.

Il est impossible de créer le vlan 2 sur le switche 1 car étant en mode client le S1 peut pas modifier quelque chose comme créer supprimer des vlans etc.

Step 7: Share the Vlan configuration of S2 with the switch S1

Cette étape vise à permettre au switch S1 de récupérer automatiquement la base de VLANs configurée sur S2. Comme S2 est configuré en mode serveur VTP, il détient la base de référence des VLANs du domaine. Pour que S1 puisse recevoir cette configuration, il doit être correctement intégré au même domaine VTP et relié à S2 via un lien trunk fonctionnel.

```
YG-S2(config-vlan)#ex
YG-S2(config-vlan)#exit
YG-S2(config)#int
YG-S2(config)#interface f0
YG-S2(config)#interface fa0/1
YG-S2(config-if)#sw
YG-S2(config-if)#switchport mo
YG-S2(config-if)#switchport mode tr
YG-S2(config-if)#switchport mode trunk
YG-S2(config-if)#sw
YG-S2(config-if)#switchport tr
YG-S2(config-if)#switchport trunk al
YG-S2(config-if)#switchport trunk allowed v
YG-S2(config-if)#switchport trunk allowed vlan 1-99
YG-S2(config-if)#ex
YG-S2(config)#ex
YG-S2#
%SYS-5-CONFIG_I: Configured from console by console

YG-S2#sh
YG-S2#show vt
YG-S2#show vtp status
VTP Version                     : 1
Configuration Revision          : 2
Maximum VLANs supported locally : 255
Number of existing VLANs        : 9
VTP Operating Mode              : Server
VTP Domain Name                 : My-office
VTP Pruning Mode                : Disabled
VTP V2 Mode                     : Disabled
VTP Traps Generation            : Disabled
MD5 digest                      : 0x30 0xE1 0x84 0x93 0xD8 0xB2 0x08 0x64
Configuration last modified by 172.17.10.2 at 3-1-93 03:26:36
Local updater ID is 172.17.10.2 on interface Vl1 (lowest numbered VLAN interface found)
YG-S2#
```

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```
YG-S3>en
YG-S3#sh
YG-S3#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/3, Fa0/4, Fa0/5, Fa0/6
                                                Fa0/7, Fa0/8, Fa0/9, Fa0/10
                                                Fa0/11, Fa0/12, Fa0/13, Fa0/14
                                                Fa0/15, Fa0/16, Fa0/17, Fa0/18
                                                Fa0/19, Fa0/20, Fa0/21, Fa0/22
                                                Fa0/23, Fa0/24
10   finance                          active
20   staff                            active
30   guest                            active
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
YG-S3#sh
YG-S3#show vtp status
VTP Version                     : 1
Configuration Revision          : 3
Maximum VLANs supported locally : 255
Number of existing VLANs        : 8
VTP Operating Mode              : Client
VTP Domain Name                 : My-office
VTP Pruning Mode                : Disabled
VTP V2 Mode                     : Disabled
VTP Traps Generation            : Disabled
MD5 digest                      : 0xA5 0x25 0xF1 0x50 0xDD 0xBF 0x60 0x5C
Configuration last modified by 172.17.10.2 at 3-2-93 12:16:32
YG-S3#
```

Copy | Past

```
YG-S1>en
YG-S1#sh
YG-S1#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/3, Fa0/4, Fa0/5, Fa0/6
                                                Fa0/7, Fa0/8, Fa0/9, Fa0/10
                                                Fa0/11, Fa0/12, Fa0/13, Fa0/14
                                                Fa0/15, Fa0/16, Fa0/17, Fa0/18
                                                Fa0/19, Fa0/20, Fa0/21, Fa0/22
                                                Fa0/23, Fa0/24
10   finance                          active
20   staff                            active
30   guest                            active
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
YG-S1#sh
YG-S1#show vtp status
VTP Version                     : 1
Configuration Revision          : 3
Maximum VLANs supported locally : 255
Number of existing VLANs        : 8
VTP Operating Mode              : Client
VTP Domain Name                 : My-office
VTP Pruning Mode                : Disabled
VTP V2 Mode                     : Disabled
VTP Traps Generation            : Disabled
MD5 digest                      : 0xA5 0x25 0xF1 0x50 0xDD 0xBF 0x60 0x5C
Configuration last modified by 172.17.10.2 at 3-2-93 12:16:32
YG-S1#
```

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```
YG-S2>en
YG-S2#sh
YG-S2#show in
YG-S2#show interfaces vl
YG-S2#show vl
YG-S2#show vlan br
YG-S2#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/21, Fa0/22, Fa0/23, Fa0/24
10   finance                          active    Fa0/16, Fa0/17, Fa0/18, Fa0/19
                                                Fa0/20
20   staff                            active    Fa0/3
30   guest                            active    Fa0/4, Fa0/5, Fa0/6, Fa0/7
                                                Fa0/8, Fa0/9, Fa0/10, Fa0/11
                                                Fa0/12, Fa0/13, Fa0/14, Fa0/15
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
YG-S2#sh
YG-S2#show vtp status
VTP Version                     : 1
Configuration Revision          : 3
Maximum VLANs supported locally : 255
Number of existing VLANs        : 8
VTP Operating Mode              : Server
VTP Domain Name                 : My-office
VTP Pruning Mode                : Disabled
VTP V2 Mode                     : Disabled
VTP Traps Generation            : Disabled
MD5 digest                      : 0xA5 0x25 0xF1 0x50 0xDD 0xBF 0x60 0x5C
Configuration last modified by 172.17.10.2 at 3-2-93 12:16:32
Local updater ID is 172.17.10.2 on interface Vl1 (lowest numbered VLAN interface found)
YG-S2#
```

Configuration of the Spanning Tree:
Le Spanning Tree Protocol (STP) est utilisé pour empêcher la formation de boucles dans les réseaux commutés comportant plusieurs chemins redondants. Il garantit qu'un seul chemin logique reste actif entre deux équipements, tout en maintenant des liens de secours qui peuvent être activés en cas de défaillance. STP contribue ainsi à la stabilité, à la résilience et à la sécurité du réseau.

```
Fa0/2            Desg FWD 19        128.2    P2p

YG-S1#show spanning-tree vlan 10
VLAN0010
  Spanning tree enabled protocol ieee
  Root ID    Priority    32778
             Address     0002.17B1.A85B
             This bridge is the root
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    32778  (priority 32768 sys-id-ext 10)
             Address     0002.17B1.A85B
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  20

Interface        Role Sts Cost      Prio.Nbr Type
---------------- ---- --- --------- -------- --------------------------------
Fa0/1            Desg FWD 19        128.1    P2p
Fa0/2            Desg FWD 19        128.2    P2p

YG-S1#show spanning-tree vlan 20
VLAN0020
  Spanning tree enabled protocol ieee
  Root ID    Priority    24596
             Address     0090.2B44.C5BE
             Cost        19
             Port        2(FastEthernet0/2)
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    32788  (priority 32768 sys-id-ext 20)
             Address     0002.17B1.A85B
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  20

Interface        Role Sts Cost      Prio.Nbr Type
---------------- ---- --- --------- -------- --------------------------------
Fa0/1            Desg FWD 19        128.1    P2p
Fa0/2            Root FWD 19        128.2    P2p

YG-S1#
```

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```
YG-S2#show spanning-tree vlan 1
VLAN0001
  Spanning tree enabled protocol ieee
  Root ID    Priority    32769
             Address     0002.17B1.A85B
             Cost        19
             Port        1(FastEthernet0/1)
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    32769  (priority 32768 sys-id-ext 1)
             Address     0090.2B44.C5BE
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  20

Interface        Role Sts Cost      Prio.Nbr Type
---------------- ---- --- --------- -------- --------------------------------
Fa0/1            Root FWD 19        128.1    P2p
Fa0/2            Altn BLK 19        128.2    P2p

YG-S2#
```

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IOS Command Line Interface

```
  Bridge ID  Priority    32770  (priority 32768 sys-id-ext 2)
             Address     0002.17B1.A85B
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  20

Interface        Role Sts Cost      Prio.Nbr Type
---------------- ---- --- --------- -------- --------------------------------
Fa0/1            Desg FWD 19        128.1    P2p
Fa0/2            Desg FWD 19        128.2    P2p

VLAN0010
  Spanning tree enabled protocol ieee
  Root ID    Priority    32778
             Address     0002.17B1.A85B
             This bridge is the root
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    32778  (priority 32768 sys-id-ext 10)
             Address     0002.17B1.A85B
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  20

Interface        Role Sts Cost      Prio.Nbr Type
---------------- ---- --- --------- -------- --------------------------------
Fa0/1            Desg FWD 19        128.1    P2p
Fa0/2            Desg FWD 19        128.2    P2p

VLAN0020
  Spanning tree enabled protocol ieee
  Root ID    Priority    32788
             Address     0002.17B1.A85B
             This bridge is the root
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    32788  (priority 32768 sys-id-ext 20)
             Address     0002.17B1.A85B
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  20

 --More--
```

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```
             Aging Time  20

Interface        Role Sts Cost      Prio.Nbr Type
---------------- ---- --- --------- -------- --------------------------------
Fa0/1            Root FWD 19        128.1    P2p
Fa0/2            Desg FWD 19        128.2    P2p

VLAN0010
  Spanning tree enabled protocol ieee
  Root ID    Priority    32778
             Address     0002.17B1.A85B
             Cost        19
             Port        1(FastEthernet0/1)
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    32778  (priority 32768 sys-id-ext 10)
             Address     0090.2B44.C5BE
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  20

Interface        Role Sts Cost      Prio.Nbr Type
---------------- ---- --- --------- -------- --------------------------------
Fa0/1            Root FWD 19        128.1    P2p
Fa0/2            Desg FWD 19        128.2    P2p

VLAN0020
  Spanning tree enabled protocol ieee
  Root ID    Priority    32788
             Address     0002.17B1.A85B
             Cost        19
             Port        1(FastEthernet0/1)
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    32788  (priority 32768 sys-id-ext 20)
             Address     0090.2B44.C5BE
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  20

Interface        Role Sts Cost      Prio.Nbr Type
---------------- ---- --- --------- -------- --------------------------------
```

```
  Spanning tree enabled protocol ieee
  Root ID    Priority    32778
             Address     0002.17B1.A85B
             Cost        19
             Port        1(FastEthernet0/1)
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    32778  (priority 32768 sys-id-ext 10)
             Address     0090.2B44.C5BE
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  20

Interface        Role Sts Cost      Prio.Nbr Type
---------------- ---- --- --------- -------- --------------------------------
Fa0/1            Root FWD 19        128.1    P2p
Fa0/2            Desg FWD 19        128.2    P2p

YG-S2#show spanning-tree vlan 30
VLAN0030
  Spanning tree enabled protocol ieee
  Root ID    Priority    32798
             Address     0002.17B1.A85B
             Cost        19
             Port        1(FastEthernet0/1)
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    32798  (priority 32768 sys-id-ext 30)
             Address     0090.2B44.C5BE
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  20

Interface        Role Sts Cost      Prio.Nbr Type
---------------- ---- --- --------- -------- --------------------------------
Fa0/1            Root FWD 19        128.1    P2p
Fa0/2            Desg FWD 19        128.2    P2p
Fa0/4            Desg FWD 19        128.4    P2p
Fa0/5            Desg FWD 19        128.5    P2p

YG-S2#
```

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Au début, chaque VLAN avait un root bridge en fonction des priorités par défaut et des adresses MAC, ce qui pouvait donner des root bridges différents selon les VLANs. Après modification manuelle de la priorité STP sur le switch 20, celui-ci a obtenu une priorité plus faible que les autres. STP l'a donc élu root bridge, et comme cette priorité était inférieure pour tous les VLANs concernés, ce même switch est devenu root bridge unique pour l'ensemble du réseau. Cela permet de stabiliser la topologie et de maîtriser le cheminement du trafic.